

Azure AutoML vs. Local Forecasting Pipeline

A head-to-head accuracy comparison across all 4 business forecasts

Report generated: July 31, 2026

Metrics compared: New Patients, Clinical Encounters, Collections, Contact Lenses

Local pipeline: 6 statistical/ML models + a backtested Ensemble

Azure AutoML: up to 25 algorithm trials per metric (statistical, ML, and blended)

Overall result: Local wins 1/4 · Azure wins 3/4 (by average accuracy)

EXECUTIVE SUMMARY

LOCAL AVG ACCURACY

88.2%

AZURE AVG ACCURACY

90.7%

METRICS WHERE
LOCAL WINS

1 / 4

METRICS WHERE
AZURE WINS

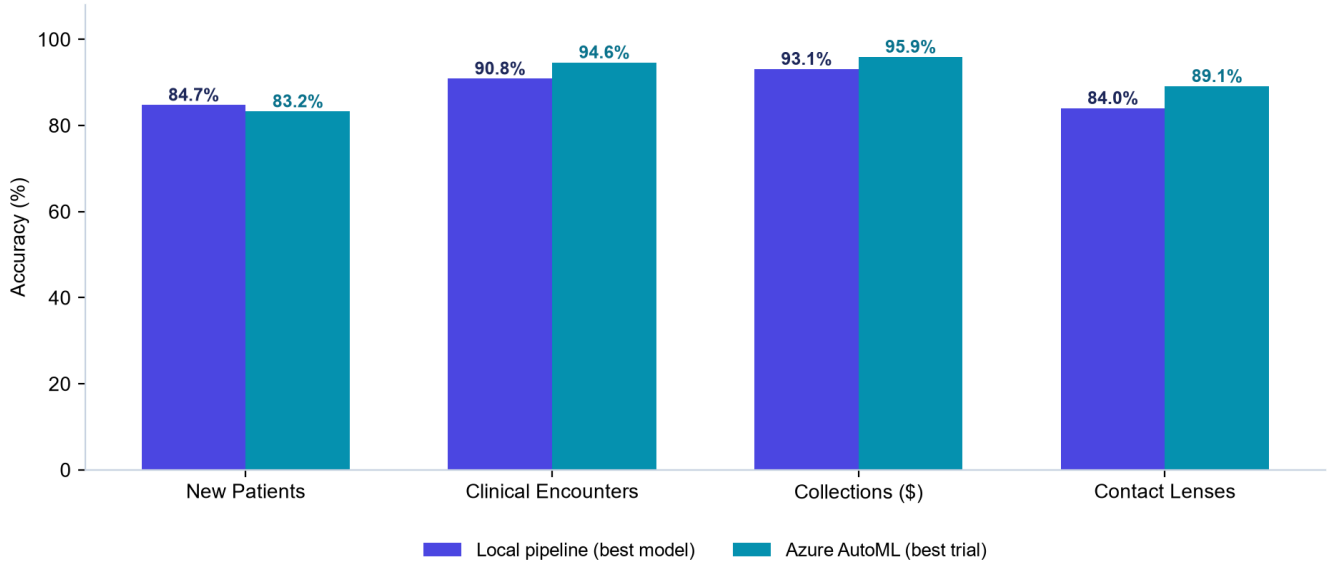
3 / 4

Both systems were tested on the exact same cleaned, future-date-filtered monthly history for all 4 business metrics. The local pipeline backtests 6 individual models (Seasonal Naive, ETS, Auto-SARIMA, Prophet, XGBoost, SARIMAX) plus a self-built Ensemble that blends the two best performers. Azure AutoML independently searched up to 25 of its own algorithms per metric — including its own automatic ensembling (**VotingEnsemble**).

Key finding: Azure AutoML's own top pick was **VotingEnsemble on all 4 metrics** — the same core idea as this project's own Ensemble model (blend the best few candidates rather than trust one). Two independent systems reached the same conclusion: **blending models beats any single model** on this kind of business data. Accuracy differences between the two systems were modest (0.6–5.1 percentage points) on every metric — this validates that the local pipeline is not leaving significant accuracy on the table, while also surfacing a few extra candidate algorithms (RandomForest, ExtremeRandomTrees) worth considering as future local additions.

Accuracy at a Glance — All 4 Metrics

Overall Accuracy: Local Pipeline vs. Azure AutoML



Metric	Local Winner	Local Accuracy	Azure Winner	Azure Accuracy	Overall Pick
New Patients	Ensemble	84.7%	VotingEnsemble	83.2%	Local
Clinical Encounters	Ensemble	90.8%	VotingEnsemble	94.6%	Azure
Collections (\$)	Ensemble	93.1%	VotingEnsemble	95.9%	Azure
Contact Lenses	ETS (Holt-Winters)	84.0%	VotingEnsemble	89.1%	Azure

Methodology Notes

- **Same data, both sides:** Azure AutoML was fed the identical already-cleaned monthly series (future dates filtered, migration/outlier months excluded) that the local pipeline trains on — not raw, unfiltered data.
- **Different accuracy formulas:** the local pipeline's `accuracy_pct` weights near-term vs. long-term forecast error and applies a flat-forecast penalty; Azure's accuracy-equivalent here is simply $100 - \text{MAPE}\%$. The two are directionally comparable but not a perfect apples-to-apples match.
- **Compute:** Azure AutoML ran on a single-node Compute Instance (Standard_NC4as_T4_v3), trying up to 25 trials sequentially per metric with 5-fold cross-validation.
- **Local pipeline:** uses rolling-origin (expanding window) backtesting — testing each model on real held-out months it never trained on, the same discipline Azure AutoML applies via cross-validation.

NEW PATIENTS

HISTORY AVAILABLE

45 mo

LOCAL WINNER

Ensemble

AZURE WINNER

VotingEnsemble

ACCURACY GAP

1.5 pts

Overall Pick for New Patients: [Local Pipeline](#)

Local pipeline's **Ensemble** reached **84.7%** backtested accuracy. Azure AutoML's best trial, **VotingEnsemble** (out of 25 algorithms tried), reached **~83.2%** accuracy-equivalent. They are effectively tied by 1.5 percentage points.

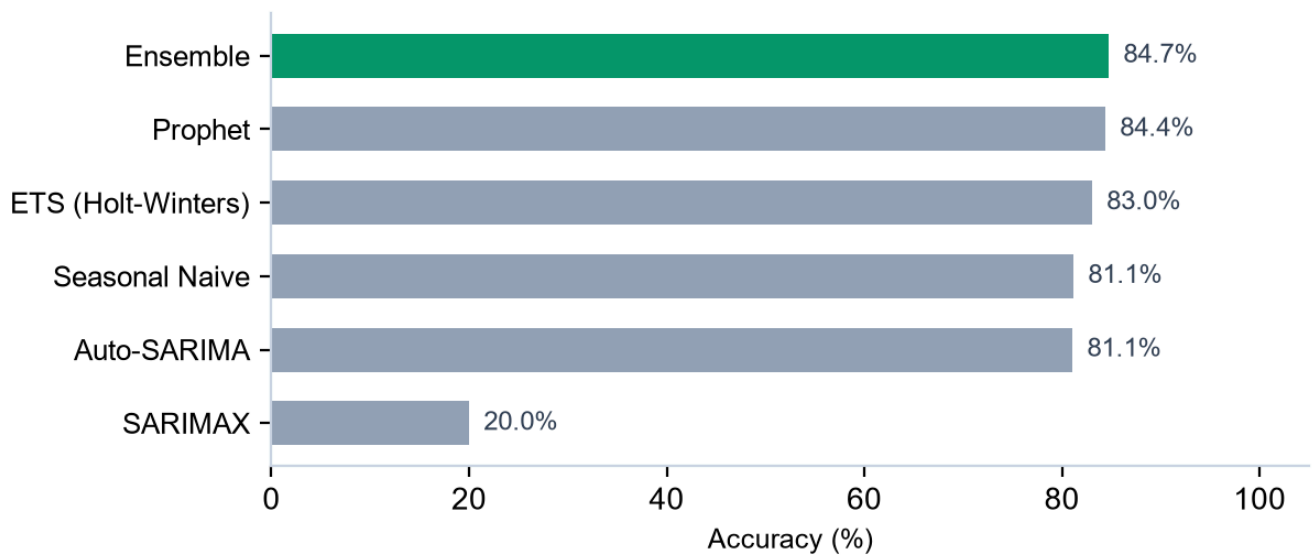
Data quality notes:

- patients: excluded Sep 2022 (82,060) as a data anomaly (robust z-score outlier, likely a migration/import artifact rather than real activity)

Local Pipeline — All 7 Candidates Backtested

Model	Backtested Accuracy	Status
Ensemble	84.7%	Selected
Prophet	84.4%	
ETS (Holt-Winters)	83.0%	
Seasonal Naive	81.1%	
Auto-SARIMA	81.1%	
SARIMAX	20.0%	

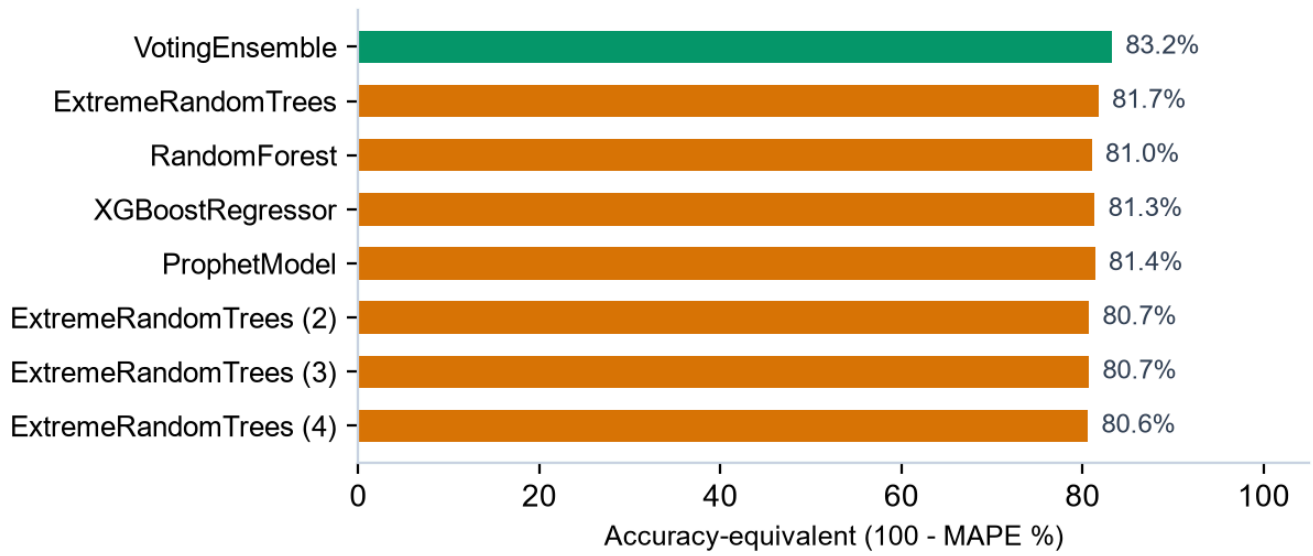
Local Pipeline — All Candidates (New Patients)



Azure AutoML — Top 10 of 25 Trials

Algorithm	Accuracy-equivalent	Normalized RMSE	MAPE
VotingEnsemble	83.2%	0.1541	16.77%
ExtremeRandomTrees	81.7%	0.1710	18.25%
RandomForest	81.0%	0.1790	19.00%
XGBoostRegressor	81.3%	0.1829	18.67%
ProphetModel	81.4%	0.1847	18.63%
ExtremeRandomTrees	80.7%	0.1850	19.31%
ExtremeRandomTrees	80.7%	0.1857	19.31%
ExtremeRandomTrees	80.6%	0.1870	19.42%
SeasonalNaive	80.5%	0.1888	19.50%
ElasticNet	79.0%	0.1989	20.96%

Azure AutoML — Top Trials (New Patients)



CLINICAL ENCOUNTERS

HISTORY AVAILABLE

161 mo

LOCAL WINNER

Ensemble

AZURE WINNER

VotingEnsemble

ACCURACY GAP

3.8 pts

Overall Pick for Clinical Encounters: [Azure AutoML](#)

Local pipeline's **Ensemble** reached **90.8%** backtested accuracy. Azure AutoML's best trial, **VotingEnsemble** (out of 25 algorithms tried), reached **~94.6%** accuracy-equivalent. Azure AutoML is ahead by 3.8 percentage points.

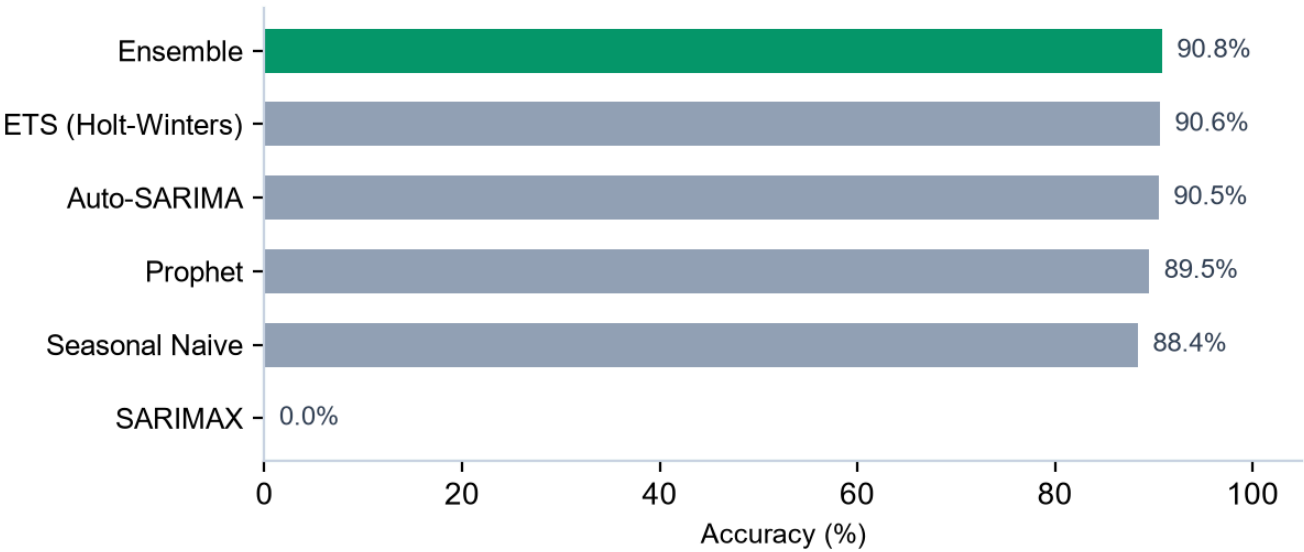
Data quality notes:

- encounters: excluded Apr 2020 (95) as a data anomaly (robust z-score outlier, likely a migration/import artifact rather than real activity)

Local Pipeline — All 7 Candidates Backtested

Model	Backtested Accuracy	Status
Ensemble	90.8%	Selected
ETS (Holt-Winters)	90.6%	
Auto-SARIMA	90.5%	
Prophet	89.5%	
Seasonal Naive	88.4%	
SARIMAX	0.0%	

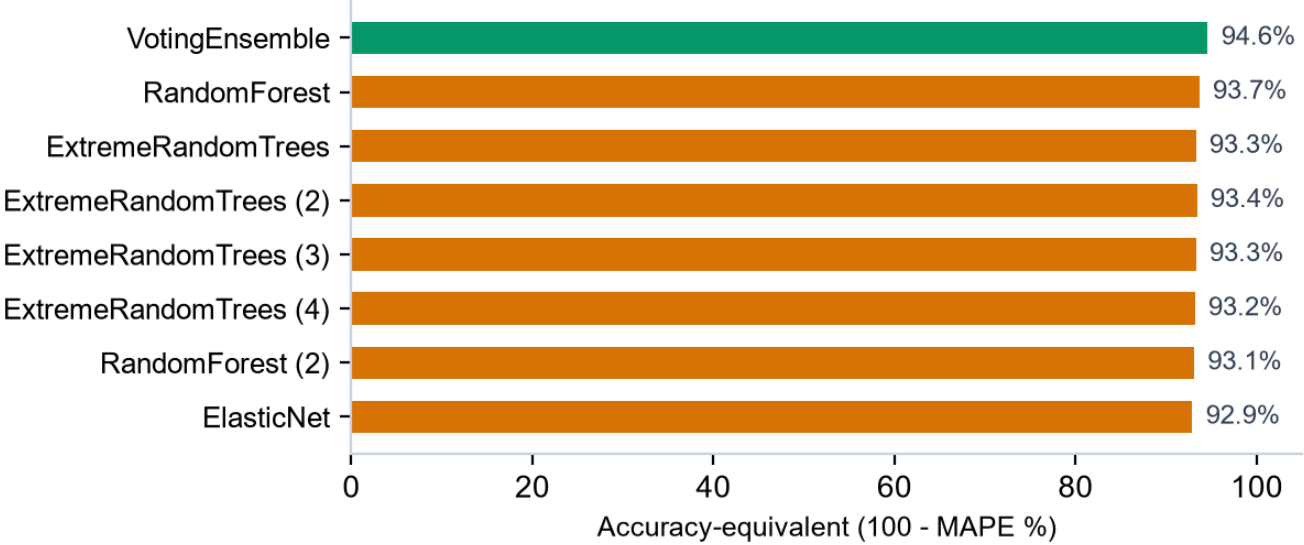
Local Pipeline — All Candidates (Clinical Encounters)



Azure AutoML — Top 10 of 25 Trials

Algorithm	Accuracy-equivalent	Normalized RMSE	MAPE
VotingEnsemble	94.6%	0.1216	5.40%
RandomForest	93.7%	0.1412	6.34%
ExtremeRandomTrees	93.3%	0.1465	6.72%
ExtremeRandomTrees	93.4%	0.1487	6.61%
ExtremeRandomTrees	93.3%	0.1543	6.71%
ExtremeRandomTrees	93.2%	0.1554	6.84%
RandomForest	93.1%	0.1579	6.91%
ElasticNet	92.9%	0.1618	7.11%
ElasticNet	92.9%	0.1619	7.12%
Average	92.4%	0.1641	7.61%

Azure AutoML — Top Trials (Clinical Encounters)



COLLECTIONS (\$)

HISTORY AVAILABLE

46 mo

LOCAL WINNER

Ensemble

AZURE WINNER

VotingEnsemble

ACCURACY GAP

2.8 pts

Overall Pick for Collections (\$): [Azure AutoML](#)

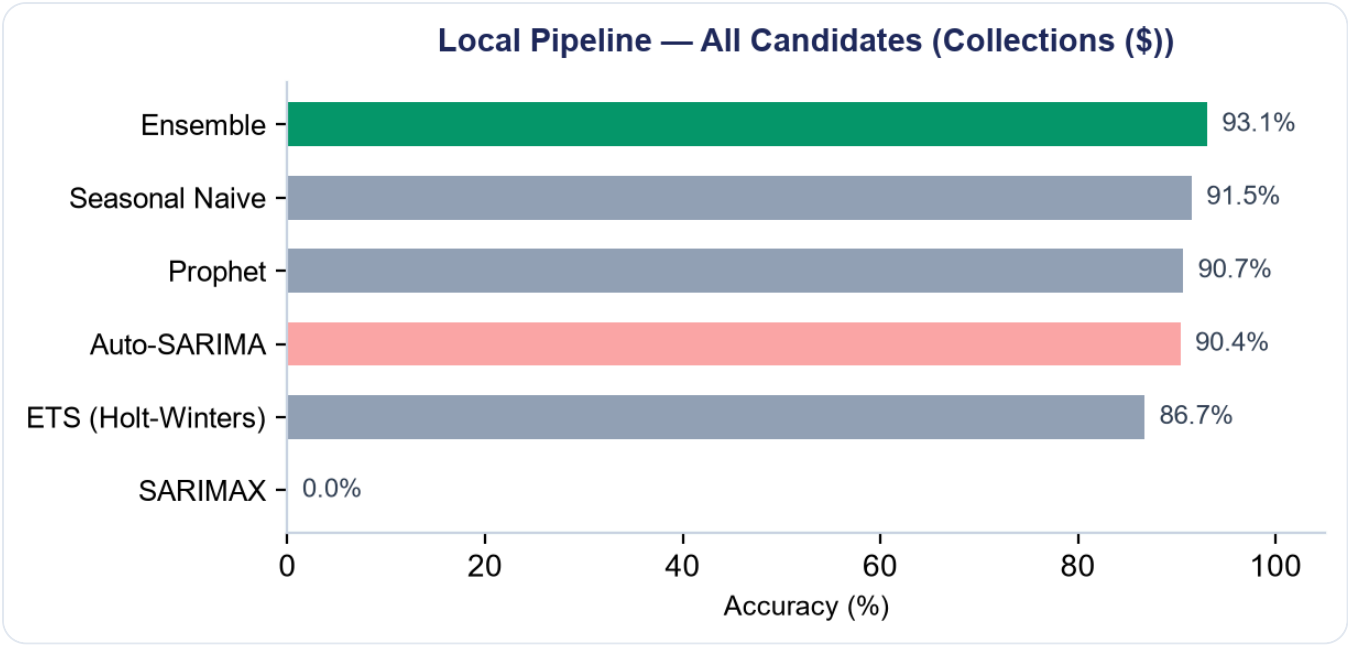
Local pipeline's **Ensemble** reached **93.1%** backtested accuracy. Azure AutoML's best trial, **VotingEnsemble** (out of 25 algorithms tried), reached **~95.9%** accuracy-equivalent. Azure AutoML is ahead by 2.8 percentage points.

Data quality notes:

- collections: excluded Oct 2018 (32) as a data anomaly (robust z-score outlier, likely a migration/import artifact rather than real activity)
- collections: excluded Jan 2020 (24) as a data anomaly (robust z-score outlier, likely a migration/import artifact rather than real activity)
- collections: excluded Jun 2020 (473) as a data anomaly (robust z-score outlier, likely a migration/import artifact rather than real activity)
- collections: excluded Jan 2021 (189) as a data anomaly (robust z-score outlier, likely a migration/import artifact rather than real activity)

Local Pipeline — All 7 Candidates Backtested

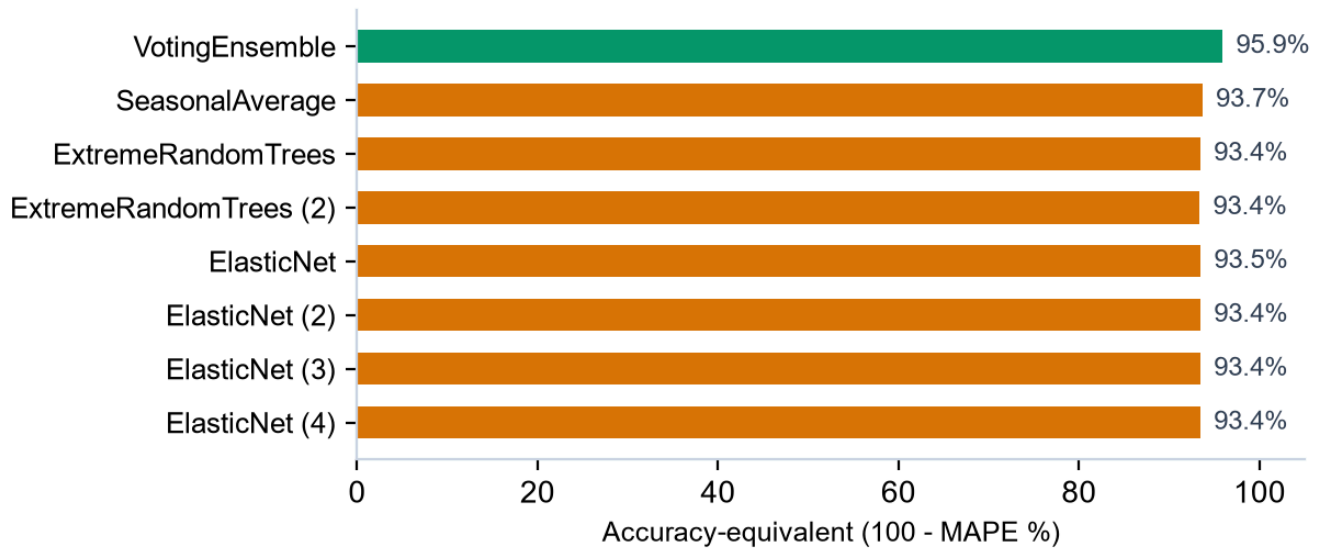
Model	Backtested Accuracy	Status
Ensemble	93.1%	Selected
Seasonal Naive	91.5%	
Prophet	90.7%	
Auto-SARIMA flat-forecast penalty	90.4%	
ETS (Holt-Winters)	86.7%	
SARIMAX	0.0%	



Azure AutoML — Top 10 of 25 Trials

Algorithm	Accuracy-equivalent	Normalized RMSE	MAPE
VotingEnsemble	95.9%	0.0814	4.10%
SeasonalAverage	93.7%	0.1345	6.34%
ExtremeRandomTrees	93.4%	0.1346	6.57%
ExtremeRandomTrees	93.4%	0.1359	6.65%
ElasticNet	93.5%	0.1373	6.54%
ElasticNet	93.4%	0.1375	6.55%
ElasticNet	93.4%	0.1378	6.57%
ElasticNet	93.4%	0.1379	6.57%
ElasticNet	93.4%	0.1379	6.57%
ElasticNet	93.4%	0.1380	6.57%

Azure AutoML — Top Trials (Collections (\$))



CONTACT LENSES

HISTORY AVAILABLE

46 mo

LOCAL WINNER

ETS (Holt-Winters)

AZURE WINNER

VotingEnsemble

ACCURACY GAP

5.1 pts

Overall Pick for Contact Lenses: [Azure AutoML](#)

Local pipeline's **ETS (Holt-Winters)** reached **84.0%** backtested accuracy. Azure AutoML's best trial, **VotingEnsemble** (out of 25 algorithms tried), reached **~89.1%** accuracy-equivalent. Azure AutoML is ahead by 5.1 percentage points.

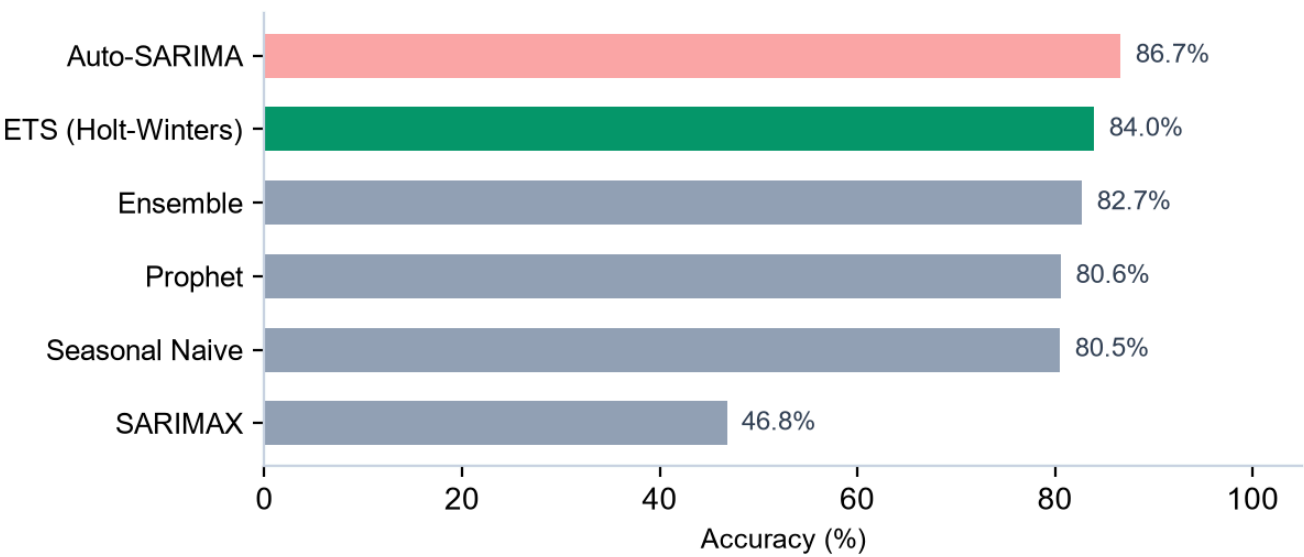
Data quality notes:

- contact lenses: excluded Jun 2020 (2) as a data anomaly (robust z-score outlier, likely a migration/import artifact rather than real activity)
- contact lenses: excluded Apr 2021 (2) as a data anomaly (robust z-score outlier, likely a migration/import artifact rather than real activity)
- contact lenses: excluded May 2021 (4) as a data anomaly (robust z-score outlier, likely a migration/import artifact rather than real activity)
- contact lenses: excluded Jul 2021 (2) as a data anomaly (robust z-score outlier, likely a migration/import artifact rather than real activity)

Local Pipeline — All 7 Candidates Backtested

Model	Backtested Accuracy	Status
Auto-SARIMA flat-forecast penalty	86.7%	
ETS (Holt-Winters)	84.0%	Selected
Ensemble	82.7%	
Prophet	80.6%	
Seasonal Naive	80.5%	
SARIMAX	46.8%	

Local Pipeline — All Candidates (Contact Lenses)



Azure AutoML — Top 10 of 25 Trials

Algorithm	Accuracy-equivalent	Normalized RMSE	MAPE
VotingEnsemble	89.1%	0.1288	10.91%
Average	89.2%	0.1311	10.81%
AutoArima	89.2%	0.1311	10.81%
LightGBM	89.2%	0.1311	10.81%
SeasonalAverage	88.6%	0.1334	11.44%
ExponentialSmoothing	88.9%	0.1341	11.05%
ElasticNet	87.7%	0.1409	12.30%
ElasticNet	87.7%	0.1410	12.30%
ElasticNet	85.5%	0.1590	14.48%
RandomForest	87.1%	0.1592	12.93%

Azure AutoML — Top Trials (Contact Lenses)

